Data Collection and Preprocessing Phase

Date	25 February 2026
Team ID	259422
Project Title	Online Payments Fraud Detection using Machine Learning
Maximum Marks	2 Marks

Data Collection Plan & Raw Data Sources Identification Template

Downloaded a large historical dataset as prescribed from Kaggle

Data Collection Plan Template

Section	Description
Project Overview	Develop a supervised machine learning–based binary classification system to detect fraudulent online payment transactions while minimizing False Negatives (Type II Errors). Given the financial and reputational cost of undetected fraud, the primary optimization goal is maximizing Recall for the Fraud class, while maintaining acceptable precision and operational efficiency.
Data Collection Plan	The data for this project will be collected from the following source: A publicly available fraud detection dataset hosted on Kaggle. Specifically, the Online Payments Fraud Detection Dataset uploaded by the Kaggle community.

Raw Data Sources Identified	Kaggle – Online Payments Fraud Detection Dataset in csv format Contains anonymized financial transaction records including transaction amount, type, balances before/after transaction, and fraud label (binary classification). Suitable for training fraud detection ML models.

Raw Data Sources Template

Source Name	Description	Location/URL	Format	Size	Access Permissions
Dataset 1	Publicly available dataset containing anonymized financial transaction records with fraud labels. Includes transaction type, amount, sender/receiver balances before and after transaction. Used for supervised fraud classification modeling.	https://www.kaggle.com/datasets/rupakroy/online-payments-fraud-detection-dataset?select=PS_20174392719_1491204439457_log.csv	CSV	470 MB	Public